Chapter 3 Testing Difference in Population Means

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Statistical Decision Making (45-752)

[1]:
```
import pandas
import matplotlib.pyplot as plt
from scipy.stats import t


# Jupyter "magic" command
# precision - round floats to 4 significant digits
%precision %.4f
```

[1]: '%.4f'

1 Testing Difference in Population Means

1.0.1 Learning objectives

After completing this chapter, you will be able to:

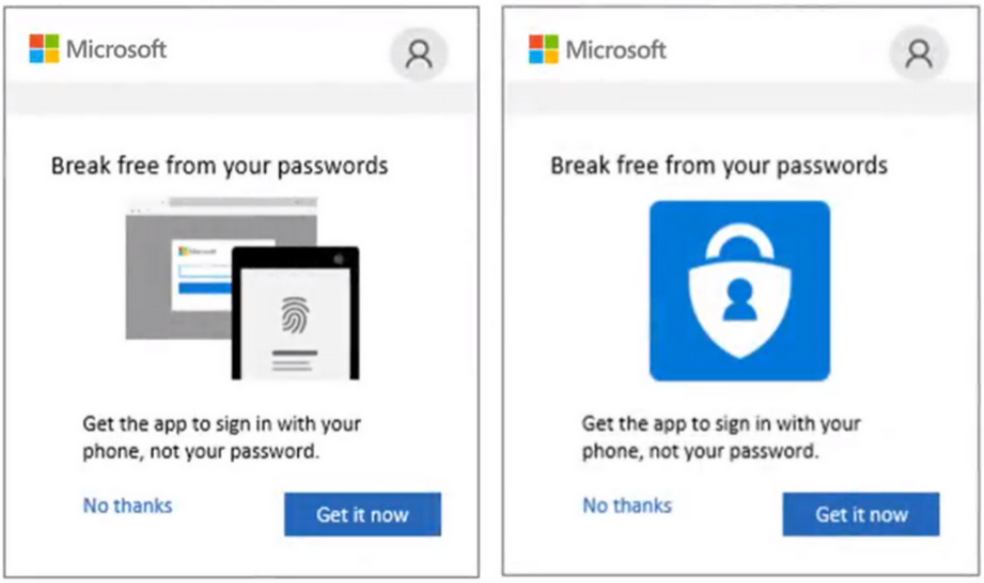
- Conduct a test of a difference in population means and interpret the result.
- Explain the purpose and logic of A/B testing in experimental design.
- Design an A/B test to evaluate competing alternatives.
- Explain why random assignment to test condition is important.

1.1 Testing A Difference in Population Means

In many situations, we want to test a difference in the means between two populations. This has come to be known as A/B testing. In this chapter, the t-statistic for testing a difference in means is introduced. Several applications are then presented to help familiarize you with conducting A/B tests.

Here is an example of an A/B test posted on the web by Microsoft.

During the A/B test one group sees the original variant and the other sees the new version (like in the A/B test by Azure Identity shown below).



The Azure Identity team ran an A/B test to see if modifying the design would affect downloads of the Authenticator app. On the left is the old variant, and on the right one of the iterations of new design

1.1.1 Refresh Your Memory about Sample Mean and Sample Variance

Suppose variable X contains the integers from 1 through 9. Calculate by hand the sample mean $\overline{X}$ and the sample variance $s_{\overline{x}}^2$. You should obtain $\overline{X} = 5$ and $s_{\overline{x}}^2 = 7.5$.

1.2 The t-statistic for The Test of Differences in Population Means

In statistics, it is common to use X and Y (rather than A and B) to denote the two alternatives being compared, with μ_x denoting the mean of population X and μ_y the mean of population Y. We wish to test a null hypothesis about a difference in population means: $(\mu_{x,0} - \mu_{y,0})$. If we have a simple random sample of size n_x from population X and a simple random sample of size n_y from population Y, the t-statistic for the test of a difference in population means is:

$$t = \frac{(\overline{X} - \overline{Y}) - (\mu_{x,0} - \mu_{y,0})}{\sqrt{\frac{s_{\overline{x}}^2}{n_x} + \frac{s_{\overline{y}}^2}{n_y}}}$$

In the above expression $\overline{X}$ and $\overline{Y}$ are the sample means, and $s_{\overline{x}}^2 = \frac{\sum_{i=1}^{n_x}(X_i - \overline{X})^2}{n_x - 1}$ and $s_{\overline{y}}^2 = \frac{\sum_{i=1}^{n_y}(Y_i - \overline{Y})^2}{n_y - 1}$ are the sample variances. The degrees of freedom for the test is $n_x + n_y - 2$.

Often, the null hypothesis is that the difference in population means equals zero: $(\mu_{x,0} - \mu_{y,0}) = 0$. The logic is the following. If this null is rejected, we then have evidence that the population means are different.

Under the null hypothesis that $(\mu_{x,0} - \mu_{y,0}) = 0$, the t-statistic for the hypothesis test then becomes:

$$t = \frac{(\overline{X} - \overline{Y})}{\sqrt{\frac{s_{\overline{x}}^2}{n_x} - \frac{s_{\overline{y}}^2}{ny}}}$$

1.3 PowerToGo Compared to ReadyPlus

The first example builds on the application to battery life from Chapter 2. Supose ReadyPlus is a competitor to PowerToGo. ReadyPlus batteries sell for slightly less than PowerToGo, but ReadyPlus claims that its batteries perform as well as PowerToGo. To investigate this claim, the manager of PowerToGo purchased 30 ReadyPlus batteries using the same randomization method that she used to select PowerToGo batteries. She obtained the battery life of each ReadyPlus battery in the same way as she did for the PowerToGo batteries. She had her assistant place the results in a variable named Hours in Excel file ReadyPlus.xlsx.

Please import the following data sets:

- PowerToGo.xlsx
- ReadyPlus.xlsx
- Battery_Data.xlsx

[2]:
```
PowerToGo = pandas.read_excel("data/PowerToGo.xlsx")
ReadyPlus = pandas.read_excel("data/ReadyPlus.xlsx")
Battery_Data = pandas.read_excel("data/Battery_Data.xlsx")
```

Let X denote PowerToGo and Y denote ReadyPlus. We calculate the sample means for the two samples using the command below. As you will see, I placed the commands to calculate the means side-by-side separated by a comma. This is simply for convenience. The two sample means will then be shown side-by side, as you see below..

[3]:
```
PowerToGo["Hours"].mean(), ReadyPlus["Hours"].mean()
```

[3]:
```
(71.3058, 68.9737)
```

Similarly, I placed the commands to calculate the sample variances side-by-side.

[4]:
```
PowerToGo["Hours"].var(), ReadyPlus["Hours"].var()
```

[4]:
```
(10.3164, 13.0757)
```

We see that mean battery life for the PowerToGo sample is $\overline{X} = 71.306$. The mean for the ReadyPlus sample is $\overline{Y} = 68.973$. The sample sizes are n_x=30 for PowerToGo and n_y=30 for ReadyPlus. The sample variance of hours for PowerToGo is $s_{\overline{x}}^2$=10.316 and the sample variance for ReadyPlus is $s_{\overline{y}}^2$=13.076.

The null hypothesis is that the population mean for ReadyPlus is the same as the population mean for PowerToGo. The manager chooses significance level α=.05. Using the results above, we calculate the t-statistic as follows. Be sure you see how the information in the previous paragraph is used in this calculation.

$$t = \frac{(\overline{X} - \overline{Y})}{\sqrt{\frac{s_{\overline{x}}^2}{n_x} - \frac{s_{\overline{y}}^2}{n_y}}} = \frac{(71.306 - 68.973)}{\sqrt{\frac{10.316}{30} + \frac{13.076}{30}}} = \frac{2.332}{.883} = 2.641$$

Using the t-statistic and information about the number of observations in each group, we calculate the p-value using the command below. The degrees of freedom is $n_x + n_y - 2 = 29 + 29 = 58$. We see that the p-value=.011. The chosen significance level is $\alpha = .05$. The p-value<α, so we reject the null hypothesis. This is good news for the manager of PowerToGo. PowerToGo batteries have significantly longer battery life than ReadyPlus.

[5]:
```
2 * t.cdf(-2.641, 58)
```

[5]:
```
0.0106
```

The data for this example are made up, but comparisons of battery life can clearly be done without great expense. If you do a web search, you will find a plethora of sites reporting comparisons of battery life. It is exceedingly rare to find a web site that reports the size of sample being used.

The web site below reported tests of three types of AA batteries: Duracell_Plus, Energizer_Max, Duracell_ Ultra

- Eight batteries of each type were tested. Is this a large enough sample for testing a difference in population means?
- While the sample size is quite small, this web site is to be commended for reporting sample size.

https://mybroadband.co.za/news/energy/244152-duracell-vs-energizer-ultimate-battery-showdown.html

1.4 Atlanta Water Conservation Experiment

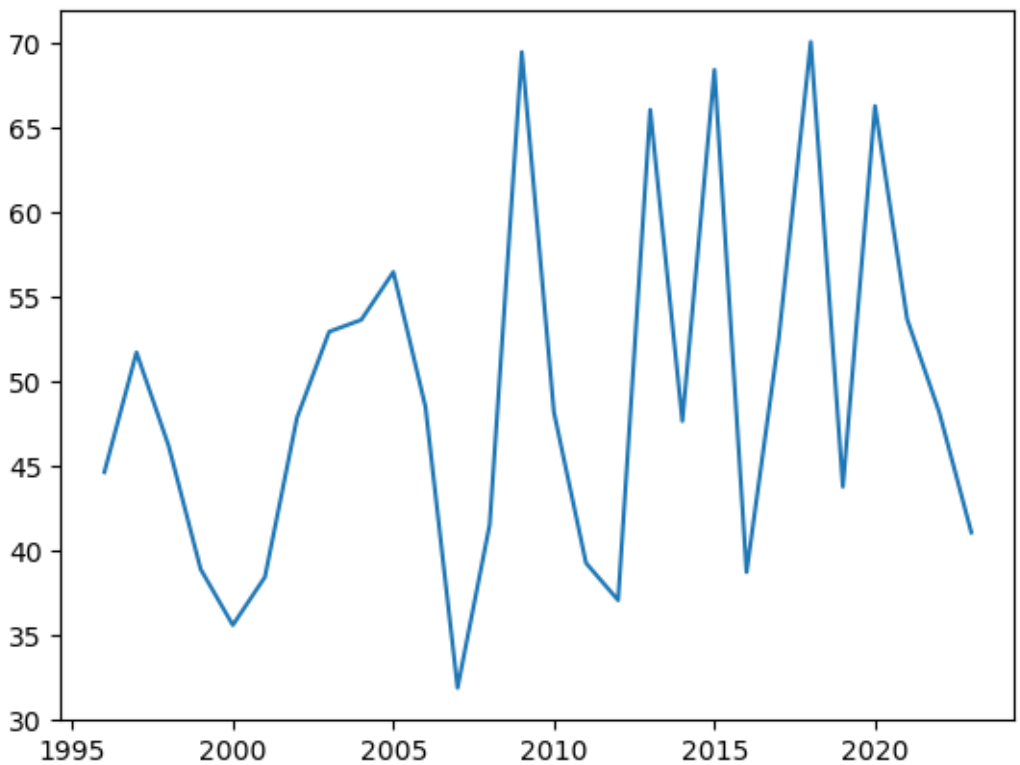
We now study an experiment to encourage water conservation. this was a remarkable large-scale experiment. Fortunately, the people who conducted the experiment shared their data with us. As you will see, we can learn a lot about statistical methods from analyzing data from the experiment. The experiment was conducted in 2007. This was was during a time when Atlanta was experiencing serious drought conditions. Import Atlanta_Rainfall_1996_2023.xlsx and give the file the shorthand name AtlantaRain.

[6]:
```
AtlantaRain = pandas.read_excel("data/Atlanta_Rainfall_1996_2023.xlsx")
```

In the file, total rainfall for each year is named Total. The command below plots the data with Year on the horizontal axis and Total rainfall on the vertical axis.

[7]:
```
plt.plot(AtlantaRain["Year"], AtlantaRain["Total"])
```

[7]:
```
[<matplotlib.lines.Line2D at 0xffff5784d3a0>]
```

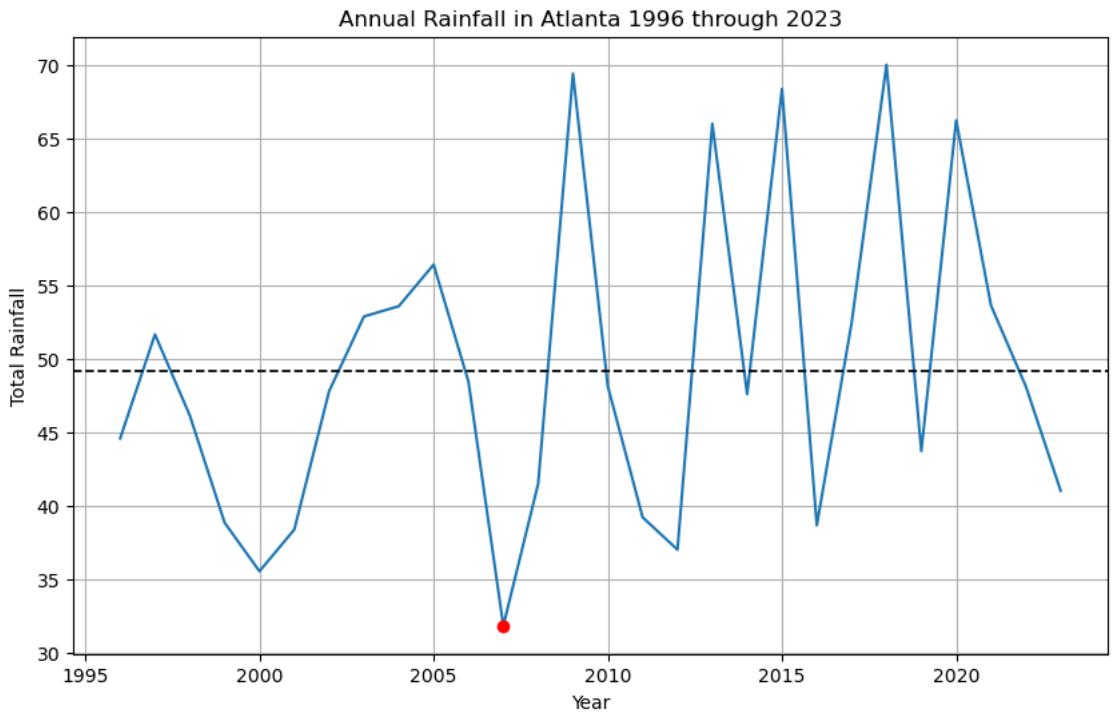


The following commands do a plot of rainfall by year from 1996 through 2023, with added detail to help the viewer. The commands show how to label the plot. To illustrate the low rainfall in year 2007, I added a red dot at (x,y) coordinates (2007,31.85). The value 31.85 for year 2007 can be found in the Total column of the Excel spreadsheet. I added a dashed horizontal line to display the average rainfall over this period of time. Then I added a grid.

You do not need to memorize these details. You can look them up here when you are doing a plot of your own.

[8]:
```
# Create the plot
plt.figure(figsize=(10, 6))
plt.plot(AtlantaRain["Year"], AtlantaRain["Total"], linewidth=1.5)
plt.xlabel("Year")
plt.ylabel("Total Rainfall")
plt.title("Annual Rainfall in Atlanta 1996 through 2023")

# Add a point for the year 2007 with rainfall of 31.85
plt.scatter(2007, 31.85, color="red", zorder=5)

# Add horizontal dashed line at y=49.20537
plt.axhline(y=49.20537, color="black", linestyle="--", linewidth=1.25)

# Add grid and make it visually clearer
```

```
plt.grid(True)

# Show the plot
plt.show()
```



The Cobb County Water System (CCWS) serves residential customers in metropolitan Atlanta, Georgia. CCWS conducted the following experiment. The experiment is the work of Paul J. Ferraro, Juan Jose Miranda, and Michael K. Price. I thank these authors for their generosity in sharing their data with us.[1]

Households were randomly assigned to one of four groups, a control group and three treatment groups.

Households in the control group did not receive any mailing.

Households in the different treatment groups received the following:

- Treatment 1: "Tip sheet", information only
- Treatment 2: Augmented "tip sheet" with personal plea to reduce usage
- Treatment 3: Explicit comparison of water usage to other Atlanta residents

Here we investigate the difference in water use between households in Treatment 3 and the control group. In a later chapter, we will investigate the other two treatments.

[1] Paul J. Ferraro, Juan Jose Miranda, and Michael K. Price, "The Persistence of Treatment Effects with Norm-Based Policy Instruments: Evidence from a Randomized Environmental Policy Experiment". *American Economic Review* vol. 101, no. 3, May 2011, (pp. 318-22).]

Each household in treatment 3 received a letter of the following form:

> As we enter the summer months, we thought that you might be interested in the following information about your water consumption last year:
> Your own total consumption June to October 2006: 52,000 gallons.
> Your neighbors' average (median) consumption June to October 2006: 35,000 gallons.
> You consumed more water than 73 percent of your Cobb County neighbors.

This is called a "Social Comparison" letter. The letters were sent at the beginning of year 2007. Data on water use were then collected for each summer from summer 2007 through summer 2013.

There were 11,675 households in Treatment group 3 and 71,642 households in the control group. Water use was measured in thousands of gallons. Mean water use for these two groups was 34.858 and 36.472. Thus, mean water use in Treatment group 3 was lower than in the control group by one thousand six hundred and fourteen gallons. This is encouraging. Was the difference significant? The sample variances for the two groups were 693.53 and 855.68 respectively.

The information in the above paragraph is all that we need to calculate the t-statistic to test the null hypothesis that there was no difference in population means. Doing the calculation, as shown below, we obtain a t-statistic of -6.04.

$$t = \frac{(\overline{X} - \overline{Y})}{\sqrt{\frac{s_x^2}{n_x} - \frac{s_y^2}{n_y}}} = \frac{(34.858 - 36.472)}{\sqrt{\frac{693.53}{11675} + \frac{855.68}{71643}}} = \frac{-1.614}{.267} = -6.04$$

The degrees of freedom is $n_x + n_y - 2 = 11675 + 71643 - 2 = 83316$

Calculating the p-value, we obtain a value barely above zero. The chosen significance level is $\alpha = .05$. Hence, we reject the null hypothesis that there was no difference in population means and conclude that the Treatment 3 letters significantly reduced water consumption.

[9]:
```
df = 11675 + 71643 - 2
2 * t.cdf(-6.04, df)
```

[9]: 0.0000

Put yourself in the position of the experimenters when they completed the above calculations. They were surely jumping for joy!

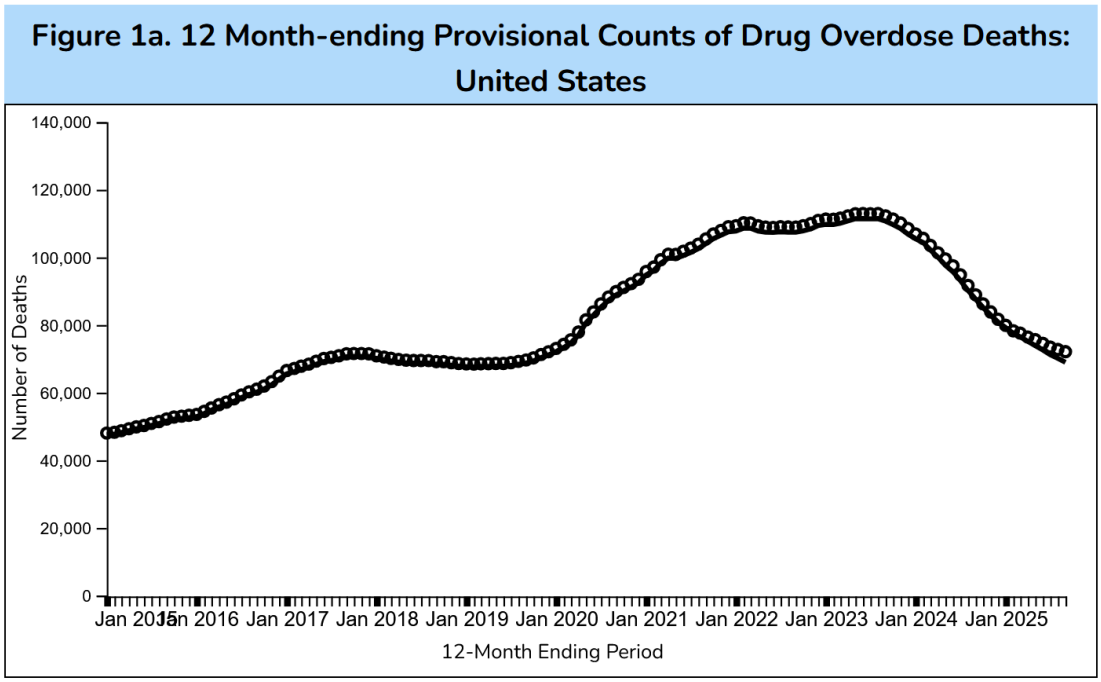
1.5 Are Opioids More Effective Than Non-opioids for Pain Relief?

News reporting of the problem of Opioid abuse declined somewhat with the emergence of Covid. Unfortunately, the scourge of opioid abuse has continued. Opioid deaths were 80,816 deaths in 2021, 110,236 in 2022, and 106,881 in 2023. [2] [3] [4] We can only hope that this recent decline continues.

[2] https://nida.nih.gov/drug-topics/trends-statistics/overdose-death-rates
[3] https://www.cdc.gov/nchs/pressroom/nchs_press_releases/2022/202205.htm
[4] https://www.cdc.gov/nchs/nvss/vsrr/drug-overdose-data.htm

While the toll on human health and welfare is the major cost of opioid abuse, the impact on business is billions of dollars:[5] [6]

The graph below shows the rapid increase in deaths from opioids since 1999.



Prescription of opioids for pain relief is a major factor leading to opioid abuse.[7] [8]

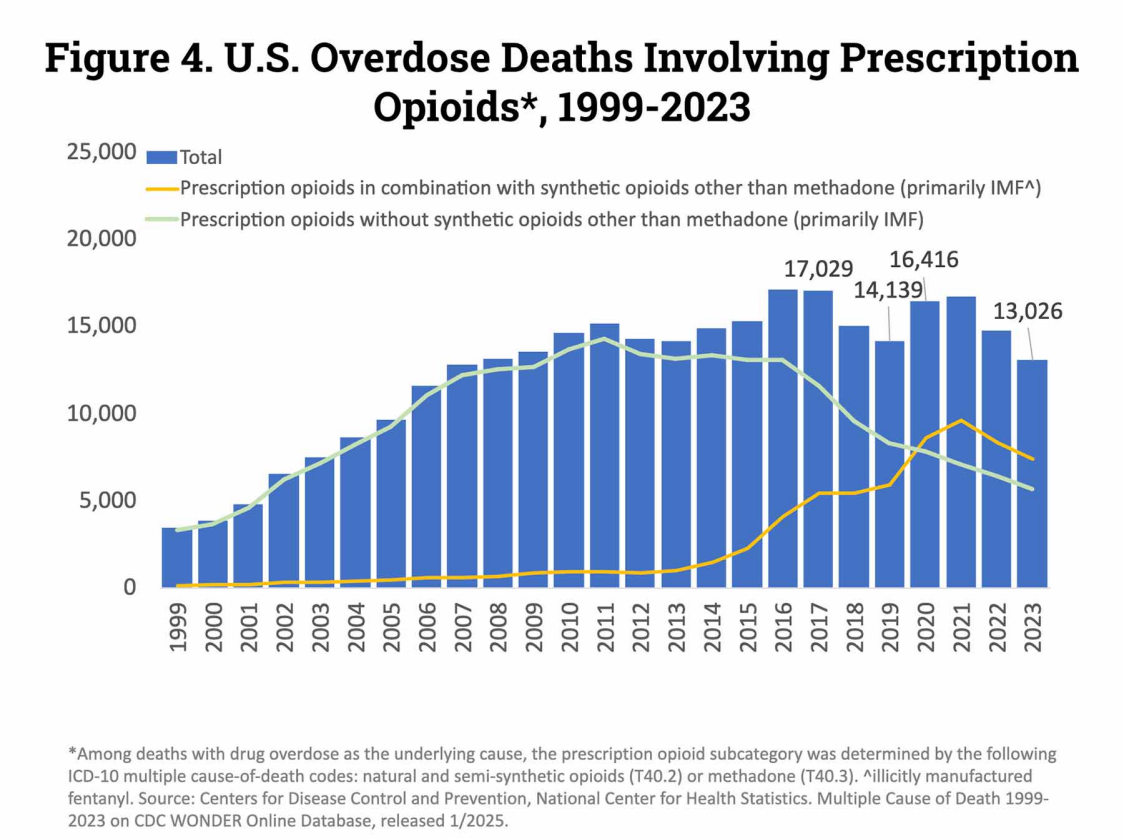
The graph below shows the more than four-fold increase in deaths from prescription opioids from 1999 to 2021.

[5] https://www.cbsnews.com/news/opioid-addiction-2-6-billion-business-cost/

[6] https://www.nytimes.com/2018/09/21/business/economy/opioid-overdose-workplace.html

[7] https://www.unodc.org/unodc/en/frontpage/2018/June/world-drug-report-2018_-opioid-crisis–prescription-drug-abuse-expands-cocaine-and-opium-hit-record-highs.html

[8] https://www.drugabuse.gov/drugs-abuse/opioids/opioid-overdose-crisis



Widespread prescription of opioids occurred because physicians were led to believe that opioids were more effective in relieving pain than available alternatives. Are opioids really more effective than safe, non-addictive alternatives for pain relief? Two experiments, detailed below, have studied this question.

On March 6, 2018 *Los Angeles Times* reported: "In the first randomized clinical trial to make a head-to-head comparison between opioids and other kinds of pain medications, patients who took opioids fared no better over the long term than patients who used safer alternatives."[9]

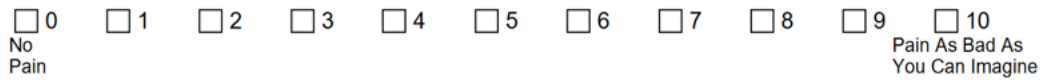
Here, we see how the experiment was undertaken, and we see how a test of differences in population means was used to evaluate the results of the experiments.

The experimenters recruited patients from Veterans Affairs primary care clinics. Eligible patients had moderate to severe chronic back pain or hip or knee osteoarthritis pain despite analgesic use. There were 238 participants. The study randomly assigned 119 of these individuals to receive opioids for their pain and 119 to receive non-opioids for their pain. In a later chapter, we will study experimental design and learn the calculations the experimenters did in deciding to to have 119 individuals in each group.

The experiment lasted for twelve months. The experimenters stated that "The primary outcome was pain-related function (Brief Pain Inventory [BPI] interference scale) over 12 months and the main secondary outcome was pain intensity (BPI severity scale). For both BPI scales (range, 0-10; higher scores=worse function or pain intensity), a 1-point improvement was clinically important."

[9] http://www.latimes.com/science/sciencenow/la-sci-sn-opioid-painkillers-no-better-20180306-story.htm

Pain intensity is measured as follows. A patient is asked to indicate pain intensity on an 11-point scale with 0 being no pain and 10 being the worst imaginable pain. The Brief Pain Inventory (BPI) scale is shown below:



Experimental outcomes were as follows. At 12 months, mean BPI pain severity was 4.0 in the opioid group (SD, 2.0) vs 3.5 in the non-opioid group (SD, 1.9). SD denotes standard deviation. Remember that lower BPI is better, so the above tells us that the non-opioid outcome was better than the opioid outcome. Is this difference significant? The information provided above permits us to do a test of differences in means using the formula presented previously in this chapter. For the non-opioid group $\overline{X} = 3.5$. For the opioid group $\overline{Y} = 4.0$. The sample sizes are n_x=119 for each group. The standard deviation is $s_{\overline{x}}^2$=1.9 for the non-opioid group $s_{\overline{y}}^2$=2.0 for the opioid group. The null hypothesis is that the population mean is the same for the two treatments. Using the results above, we calculate the t-statistic as follows:

$$t = \frac{(\overline{X} - \overline{Y})}{\sqrt{\frac{s_{\overline{x}}^2}{n_x} - \frac{s_{\overline{y}}^2}{ny}}} = \frac{(3.5 - 4.0)}{\sqrt{\frac{1.9^2}{119} + \frac{2^2}{119}}} = \frac{-.5}{.283} = -1.98$$

We have 119 individuals in each group. Hence, the degrees of freedom is $2 * 119 - 2 = 236$. Calculating the p-value, we obtain .049. This p-value is less that α=.05.

[10]:
```
2 * t.cdf(-1.98, 236)
```

[10]: 0.0489

Thus, we reject the null hypothesis that the two were equally effective in reducing pain severity and conclude that non-opioids were significantly **more** effective than opioids in reducing pain severity! This application shows the tremendous value of a well-designed experiment coupled with careful statistical analysis. Another experiment on opioids for pain relief was conducted in hospital emergency rooms. The study found that an ibuprofen-Tylenol combination was as effective as opioids.[10]

Thus, two experiments on two very different types of pain, chronic pain and pain due to injuries, find that opioids are no better than non-opioids.

1.5.1 Several Companies Played a Role in Propagating Opioid Use

Purdue Pharma played an appalling role in marketing opioids, surely one of the most egregious violations of business ethics in the history of the US. On September 17, 2021, a *New York Times* article carried the headline: "Purdue Pharma Is Dissolved and Sacklers Pay \$4.5 Billion to Settle Opioid Claims." As reported by the Times, "Purdue Pharma, the maker of the highly addictive painkiller OxyContin, was dissolved on Wednesday in a wide-ranging bankruptcy settlement that will require the company's owners, members of the Sackler family, to turn over billions of dollars of

[10] https://www.nytimes.com/2017/11/08/well/live/alternatives-to-opioids-for-pain-relief.html

their fortune to address the deadly opioid epidemic." The article goes on to say "Purdue pleaded guilty to federal criminal charges for drastically downplaying OxyContin's addictive properties and, years later, for soliciting high-volume prescribers."[11]

Purdue Pharma was not alone in propagating opioid use. On November 22, 2022, the *Wall Street Journal* reported that CVS and Wahlgreens had agreed to pay more than \$10 billion in a landmark settlement to resolve opioid-crisis lawsuits brought by states, cities and other governments. The article also reported "In 2021, a group of states reached a roughly \$25 billion settlement with Johnson & Johnson and drug distributors including McKesson Corp., Amerisource Bergen Corp. and Cardinal Health Inc. The deal was completed this summer."[12]

There is an obvious lesson in these accounts of managerial malfeasance. Sacrificing patient health and well being for financial gain is unconscionable. [13]

1.6 Why Randomize When Conducting an Experiment?

Suppose an experiment is conducted as follows. Each of 238 subjects is asked to complete the Brief Pain Inventory. The subjects are then ordered from highest to lowest pain. The 119 subjects with high pain receive opioids. The remaining 119 subjects receive non-opioids. Would this be an acceptable design of an experiment?

The answer is obviously no. For subjects with high pain there is more scope for pain reduction than for subjects with low pain. Thus, an experiment as just described would stack the deck in favor of opioids.

When we conduct an experiment comparing alternatives, we seek to establish causality. With random assignment, a subject with high pain is equally likely to be assigned to each group. The same is true for a subject with low pain. Indeed, the same is true for all subjects, regardless of their pain level. Hence, if we find a significant difference in pain reduction in one group relative to the other, we have evidence that the difference was caused by differences in effectiveness of the two medications. We have evidence of causality.

An experiment using random assignment to compare alternatives is called a **Randomized Control Trial**. Such experiments are often referred to by the shorthand **RCT**. Please be sure to remember this terminology. You might be asked in an interview to explain the meaning of RCT. You might also be asked to explain how and why RCT's are conducted.

1.7 Double-blind Experiments

When an experiment is conducted, it is important that the subjects do not know which alternative they have received. Why? If subjects believe opioids are more effective than non-opioids, their assessment of effectiveness may be biased by their beliefs.

It is also important that the person dispensing the medication does not know which treatment a subject is receiving. By way of illustration, suppose the experiment involves comparing two medications. Suppose the medications are in the form of pills. The pills for the two alternative treatments are produced so that they look identical. Let's denote the two medications by A and B. The person conducting the experiment assigns a number to each subject. Then, subjects are

[11] https://www.nytimes.com/2021/09/01/health/purdue-sacklers-opioids-settlement.html
[12] https://www.wsj.com/articles/cvs-to-pay-5-billion-to-settle-opioid-lawsuits-11667358371?mod=djem10point
[13] By the way, Purdue Pharma has never had any affiliation with Purdue University.]

randomly assigned to treatment. If a subject is assigned to treatment A, the person conducting the experiment, puts the subject's number on a bottle of pills containing treatment A. If a subject is assigned to treatment B, the person conducting the experiment, puts the subject's number on a bottle of pills containing treatment B. The person conducting the experiment then gives the bottles to the person dispensing the medication. That person doesn't know which medication is in a bottle, so the person cannot inadvertently let the subjects know which treatment they are receiving. This is called **double-blind** because the person dispensing the medication does not know which medication is in the bottle and the person receiving the medication does not know which medication is in the bottle.

If possible, keep subjects from interacting with each other. Why? Here is an example. Some illness may cause permanent disability or even death. Consider such an illness. Suppose a new treatment has been developed that has promise of being superior to the best existing alternative. The experiment is conducted to determine whether the medication is indeed more effective.

Consider the subjects in the experiment. Given the severity of the illness, most will want the new medication. Suppose for concreteness, each subject receives 30 pills, and is to take one pill a day. Refills are provided to each subject every 30 days for the duration of the experiment. Subjects don't know which medication they are receiving, but they know that the new medication is thought to be superior to the other. They want the new medication. Every 30 days after receiving their new bottle of pills, all subjects meet when the person conducting the experiment is not present. The subjects all place their pills into a large bowl. They stir the bowl thoroughly. Each then takes out 30 pills. Why would subjects do this? Each subject can expect to receive both types of medication. Given the severity of the illness they have, they know that they face severe disability or death, so they have an incentive to try to get some of the new medication.

The preceding is not a fictional story. I learned of an experiment in which such sharing of medications occurred. Obviously, this rendered the experiment invalid. Where there is risk that this kind of interaction among subjects might occur, subjects can be asked to agree to monitoring to assure they do not to meet with each other. For example, the person conducting the experiment might require participants to agree to having their locations monitored; each subject might be required to wear a location monitor that cannot be removed. Any subjects who come together must be removed from the experiment, with the medication discontinued. Knowing this, subjects will avoid getting together with each other.

Pharmaceutical companies spend a great deal in developing and obtaining approval of new drugs. Clinical trials are essential to the approval process. You may find it of interest to learn more about this industry and the role of clinical trials. If so, the following document is a good place to start. I emphasize that reading this is optional and not subject matter for a quiz or exam.

The US Congressional Budget recently issued a report titled *Research and Development in the Pharmaceutical Industry*. If you would like a copy of the report, go to the following web page: https://www.cbo.gov/publication/57126 On the right-hand-side of the page that opens, click on "Download Document" and you will receive a pdf of this 25-page document. You might also find it interesting to read about the history of clinical trials. The following document states that the US Food and Drug Administration began in 1961 to employ its authority to regulate clinical trials and clinical trial methodology: https://www.fda.gov/media/110437/download. The document also quotes a study reporting that only 45% of clinical trials in 1951 had a control group.

This preceding document also reports on the costs of clinical trials and the successs rate of medications tested: "Developing new drugs is a costly and uncertain process, and many potential drugs

never make it to market. Only about 12 percent of drugs entering clinical trials are ultimately approved for introduction by the FDA. In recent studies, estimates of the average R&D cost per new drug range from less than $1 billion to more than $2 billion per drug."

The above paragraph gives a striking example of the role of experiments in practice. It also highlight the great challenges in creating new drugs that, when subjected to experimental testing, are effective and do not have severe side effects.

1.8 Summary

This chapter has presented the approach to testing differences in means, including examples to illustrate the value of the approach. Please review the learning objectives at the beginning of this chapter to be sure you have a good understanding of the approach.